

Alternatives in Conditional Modality

Dean McHugh

MIT LingLunch, March 5, 2026

The standard theory of modals

Kripke (1959), Hanson (1965), Kratzer (1977, 1981)

- *Possibly* A ($\Diamond A$)
 A is true at **some** world in the relevant domain.
- *Necessarily* A ($\Box A$)
 A is true at **every** world in the relevant domain.

Challenges for the standard theory

Challenge 1. Free choice

- (1) Juan may take syntax or semantics. $\Diamond(A \vee B)$
a. $\rightsquigarrow$ Juan may take syntax. $\Diamond A$
b. $\rightsquigarrow$ Juan may take semantics. $\Diamond B$
- (2) Juan might take syntax or semantics. $\Diamond(A \vee B)$
a. $\rightsquigarrow$ Juan might take syntax. $\Diamond A$
b. $\rightsquigarrow$ Juan might take semantics. $\Diamond B$

Ross (1941), von Wright (1968), Kamp (1973)

' $\rightsquigarrow$ ' denotes an intuitive inference, remaining neutral on its precise status.

The standard theory does not predict these inferences, just like (3).

- (3) a. Juan may visit the department.
 $\rightsquigarrow$ Juan may visit the department and burn it down.
b. Juan may give a talk.
 $\rightsquigarrow$ Juan may give a five hour talk.

Challenge 2. Independence inferences

- (4) *Canasta*. The player has two options: (i) take all of the cards in the discard pile; (ii) take none of them.
You may take any cards in the discard pile. $\Diamond \exists x(Fx)$
- (5) There are two cards left in the discard pile, A and B .
You may take A or B . $\Diamond(A \vee B)$

Menéndez-Benito (2005), Menéndez-Benito (2010), Booth (2022)

Menéndez-Benito (2010)

Booth (2022)

Incorporating free choice (4) and (5) are equivalent, respectively, to

- (6) For every collection of cards, you may take it. $\forall x \Diamond Fx$
- (7) You may take A and you may take B . $\Diamond A \wedge \Diamond B$

Problem: the standard theory predicts (6) and (7) to be true.

Reply: Exclusive readings.

For each collection of cards, there is a deontically accessible world where 'You take that collection' is true (namely, a world where you take every card).

For each card A and B , there is a deontically accessible world where 'You take A ' is true and one where 'You take B ' is true (namely, a world where you take both cards).

(4) $\equiv$ For each collection of cards in the discard pile, you may take those *and only those* cards.

(5) $\equiv$ You may take *only* A (A *and not* B) or *only* B (B *and not* A).

Challenges for the exclusive reply.

Sometimes the exclusive inference is contradictory.

(8) Any bishop may meet a bishop.

Szabolcsi (2019), adapting an example of Elbourne's (2002).

If the exclusive inference is pragmatic, we expect it to be defeasible.

Idea: the exclusive inference does not arise when it results in a contradictory meaning.

See e.g. Alsop (2024)

(9) *A person is 'engraced' just in case they are in the presence of another bishop or bishops. Suppose that bishops are permitted to meet all together or not at all (to prevent schisms).*

Any bishop may be engraced.

(10) *Alice is playing a game with tokens. Figure (10) shows the current board, with two empty squares. The permitted next moves are to either (i) do nothing, or (ii) move the left token to the left and the right token to the right.*

Any token may have fewer neighbours than it currently has.

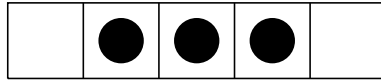


Figure 1: The current board.

(10) seems to permit Alice to move only the left token to the left, leaving the right token where it is.

The exclusive reading is **contradictory**.

(11) For every token x , x *and only* x may have fewer neighbours than it currently has.

On the non-exclusive, free-choice reading, the standard theory of modals predicts (10) to be **true**.

(12) For every token x , there is a deontically accessible world where x has fewer neighbours than it currently has.

Similar concerns arise for disjunction.

(13) *An immigration law states, 'In this document, "Ireland" refers to the island consisting of the Republic of Ireland and Northern Ireland, and "the United Kingdom" refers to the nation consisting of England, Scotland, Wales, and Northern Ireland'. The law later states:*

If you hold an E visa, you may travel to Ireland or the United

Kingdom.

Intuitively, (13) permits those with an E visa to travel to Northern Ireland.

On an exclusive reading, (13) states:

- (14) If you hold an E visa, you may visit Ireland but not the United Kingdom, or the United Kingdom but not Ireland.

(14) clearly does **not** permit one to visit Northern Ireland. It is silent on whether one may go there.

Challenge 3. Substitution of logically equivalent sentences

The standard theory predicts logically equivalent sentences to have the same modal status, e.g. both are permitted, or neither are.

- (15) *Each customer may buy at most five tickets to the concert.*
 a. Alice may buy more than four tickets.
 b. Alice may buy five or more tickets.

A case from Fine (2014):

- (16) *Eve finds herself in Alternative Eden, with one forbidden apple a_0 and infinitely many safe apples $a_1, a_2, \dots$.*
 a. Eve may eat infinitely many of the apples $a_1, a_2, \dots$.
 b. Eve may eat infinitely many of the apples $a_0, a_1, a_2, \dots$.

The prejacentes are logically equivalent: Eve eats infinitely many of the apples $a_1, a_2, \dots$ if and only if she eats infinitely many of the apples $a_0, a_1, a_2, \dots$.

- (17) *The forbidden apple is red and the safe apples are green.*
 a. Eve may eat infinitely many of the green apples.
 b. Eve may eat infinitely many of the apples.

Fine endorses Universal Realisability for Permission: a statement is permitted just in case *every way* for it to obtain is permitted. In truth-maker semantics (Fine 2017), these ‘ways’ for a sentence to obtain are its exact verifiers.

Problem: Some sentences that differ in their modal status seem to have the same exact verifiers.

- (18) a. Alice buys more than four tickets.
 b. Alice buys five or more tickets.

It is plausible that (18a) and (18b) have the same exact verifiers;

I do not rule out that (13) may also have an exclusive reading. What is nonetheless clear is that the judgement that (5) is unacceptable is far more robust than the judgement that (13) does not permit one to visit Northern Ireland. The discrepancy calls for explanation.

This example is based on one by Fox (2007:103), who judges it acceptable to say, ‘John is allowed to smoke more than six cigarettes. More specifically, he is allowed to smoke seven but not more than seven’. I slightly modified the example since *smoking cigarettes* is arguably not discrete—one can smoke half a cigarette. If John smoked six and a half cigarettes, then he smoked more than six cigarettes but did not smoke seven or more: the expressions are no longer equivalent. But one can’t buy four and a half tickets.

A single apple can never be the difference between a finite and infinite set.

namely, the state of Alice buying five tickets, the state of her buying six tickets, and so on. But they come apart under permission.

Permission is *coarse*, something can be permitted even when some ways for it to obtain are impermissible, such as (15a). Further evidence for coarseness:

- (19) *Alice and Bob are colleagues in Quebec. They know both English and French. Bill 96 requires them to speak French at work, but they defy the law by talking to each other at work in English.*
Alice and Bob are allowed to talk to each other at work.
- (20) *There are two switches, A and B. Each switch can be either up or down. The rules specify that A must be up and that B may be in any position.*
The positions of the switches are allowed to agree.

(19) and (20) are intuitively true. However, some ways for Alice and Bob to talk to each other (English) are forbidden, and some ways for the positions of the switches to agree (both down) are forbidden.

Conditional Modality

May I open the window?

- Is there at least one allowed case in which I open the window? ✗
- Would it be OK if I opened the window? ✓

Leibniz: what is permitted is ‘what is possible for a good person to do’, what is obligatory is ‘what is necessary for a good person to do’.

Hilpinen (2017:159–60) suggests paraphrasing Leibniz’s thought here using conditionals:

- One is permitted to do something just in case *if they do it*, possibly, they are a good person.
- One is obligated to do something just in case *if they do not do it*, necessarily, they are not a good person.

Kanger (1957): *ought A* says that what morality prescribes strictly implies *A*, i.e. $\Box(\neg A \supset \neg Q)$.

Anderson (1958:103): ‘to say that *p* is obligatory is to say that failure of *p* leads to a state-of-affairs [...] which is “bad”’. ‘It is *analytic* of the notion of obligation that if an obligation is not fulfilled, then something has gone wrong’ (Anderson 1967:346–47).

This is remarkably close to how a number of languages typically express permission and obligation, such as Japanese, Korean, and Burmese. (21) and (22), from Akatsuka (1992), show this for Japanese.

Leibniz to Arnauld, 1671; *Sämtliche Schriften und Briefe* II i 173–174; *Philosophical Papers and Letters*, 150.

$\Box$ ‘is the notion of analytic necessity’, *Q* ‘a constant stating what morality prescribes’, and $\supset$ the material conditional.

The negation of this ‘bad’ state, that all obligations have been met, has also been called ‘the good thing’ by Lokhorst (2006), Barker (2010), and ‘OK’ by Asher and Bonevac (2005).

Kaufmann (2017) calls these *conditional evaluative constructions*. For discussion, see Akatsuka (1992), Wymann (1996), Nauze (2008), Knoob (2008), Narrog (2009), Kaufmann (2017), Chung (2019), Li (2025).

- (21) *Permission:*
 Tabe-temo ii.
 eat even if good
 Literally: “It is good even if you eat” = “You may eat”
- (22) *Obligation:*
 Tabenakere-ba ikenai/ dame da.
 eat Neg if can go Neg no good is
 Literally: “It is not good if you don’t eat” = “You must eat”

Williamson (2007:157, 297) also considers analysing metaphysical possibility and necessity in terms of counterfactuals. A statement is metaphysically necessary just in case were it false, a contradiction would obtain, with possibility the dual of necessity.

A simple implementation

Let f be a conditional selection function, taking (at least) a proposition and a world and returning a set of worlds.

- $A \Box \rightarrow C$ is true at w iff C is true at **every** world in $f([A], w)$
 $A \Diamond \rightarrow C$ is true at w iff C is true at **some** world in $f([A], w)$

We add a constant ‘OK’ stating that the relevant ideals (deontic/epistemic) are met.

- (23) a. *Possibly A* means: if A were true, there would be some case where the relevant ideals are met. $A \Diamond \rightarrow \text{OK}$
 b. *Necessarily A* means: if A were false, there would be no case where the relevant ideals are met. $\neg A \Box \rightarrow \neg \text{OK}$

We may add constraints on f —such as *success*: A is true at every world in $f([A], w)$ —as needed.

To account for the differences between circumstantial and epistemic modals, we may introduce a circumstantial selection function f_c and an epistemic selection function f_e , and different notions of ‘OK’.

Reducing Free Choice to Simplification of disjunctive antecedents

Simplification of disjunctive antecedents (SDA):

The inference from *if A or B, C* to *if A, C* and *if B, C*.

A number of authors aim to reduce Free Choice inferences to SDA.

- The strict conditional $\Box(A \supset C)$ validates SDA
- The variably strict conditional $A \Box \rightarrow C$, with additional constraints, validates SDA.
- The existential conditional $A \Diamond \rightarrow C$ clearly does **not** validate SDA.

Is permission strong or weak?

- Strong permission: $A \Box \rightarrow \text{OK}$
- Weak permission: $A \Diamond \rightarrow \text{OK}$

Above we saw evidence that permission is coarse—(15a), (19), (20)—which are evidence for weak permission over strong permission.

Including Hilpinen (1982), Asher and Bonevac (2005), van Rooij (2006), Klinedinst (2007), Barker (2010), Franke (2011), Willer (2018), Li (2025).

Though Alonso-Ovalle (2006:18–19) observes that *might* counterfactuals also have simplification readings.

von Wright (1968), Hilpinen (1982), Li (2025)
 Anderson (1956)

Alternatives

Conditional antecedents are interpreted via sets of propositions, called their ‘alternatives’. (Alonso-Ovalle 2006, 2009, Ciardelli 2016, Santorio 2018, Willer 2018, Khoo 2022). A conditional statement holds just in case it holds for each of its alternatives.

For instance, a disjunctive conditional antecedent *if A or B* may have as alternatives the individual disjuncts *A* and *B*.

I will implement this in a first-order inquisitive semantics.

- Each sentence *A* denotes a set of states $[A]$.
- A *state* is a set of possible worlds.
- A sentence’s *alternatives* are the maximal states in its denotation.

$$\text{alt}(A) = \{s \in [A] : s \subsetneq t \text{ for no } t \in [A]\}$$

Approach from Ciardelli (2016):

- $A \Box \rightarrow C$ is true at w iff $\forall p \in \text{alt}(A) \exists q \in \text{alt}(C) : f(p, w) \subseteq q$
- $A \Diamond \rightarrow C$ is true at w iff $\forall p \in \text{alt}(A) \exists q \in \text{alt}(C) : f(p, w) \cap q \neq \emptyset$

Key result: this guarantees free choice even for $\Diamond \rightarrow$:

$$(A \vee B) \Diamond \rightarrow C \models (A \Diamond \rightarrow C) \wedge (B \Diamond \rightarrow C)$$

$$\text{A special case: } (A \vee B) \Diamond \rightarrow \text{OK} \models (A \Diamond \rightarrow \text{OK}) \wedge (B \Diamond \rightarrow \text{OK})$$

$$\text{Free choice: } \Diamond(A \vee B) \models \Diamond A \wedge \Diamond B$$

Homogeneous alternatives

Add a **homogeneity presupposition** over alternatives:

- $A \Box \rightarrow C$ presupposes at w that $\forall p, p' \in \text{alt}(A)$,
($\exists q \in \text{alt}(C) : f(p, w) \subseteq q$) iff ($\exists q \in \text{alt}(C) : f(p', w) \subseteq q$)
- $A \Diamond \rightarrow C$ presupposes at w that $\forall p, p' \in \text{alt}(A)$,
($\exists q \in \text{alt}(C) : f(p, w) \cap q \neq \emptyset$) iff ($\exists q \in \text{alt}(C) : f(p', w) \cap q \neq \emptyset$)

- (24) None of my friends would have fun at the party if Alice or Bob went. $\neg \exists x((A \vee B) \Box \rightarrow Cx)$

- a. $\rightsquigarrow$ None of my friends would have fun at the party if Alice went. $\neg \exists x(A \Box \rightarrow Cx)$

- b. $\rightsquigarrow$ None of my friends would have fun at the party if Bob went. $\neg \exists x(B \Box \rightarrow Cx)$

- (25) You may not have ice cream or cake. $\neg \Diamond(A \vee B)$

- a. You may not have ice cream. $\neg \Diamond A$

- b. You may not have cake. $\neg \Diamond B$

Clauses from Ciardelli et al. (2018:59–61):

$$|R(t_1, \dots, t_n)| = \{w \in W \mid \langle I_w(t_1), \dots, I_w(t_n) \rangle \in I_w(R)\}$$

We assume that for any element d of the domain, the language contains an individual constant d' which is always interpreted as d .

- $[R(t_1, \dots, t_n)] := \mathcal{P}(|R(t_1, \dots, t_n)|)$
- $[\neg A] := [A]^*$
- $[A \wedge B] := [A] \cap [B]$
- $[A \vee B] := [A] \cup [B]$
- $[A \rightarrow B] := [A] \Rightarrow [B]$
- $[\forall x.A(x)] := \bigcap_{d \in D} [A(d')]$
- $[\exists x.A(x)] := \bigcup_{d \in D} [A(d')]$
- $[!A] := [A] \cup [A]^*$

Santorio (2018:524)

Probability operators

Santorio (2021:633)

Alice threw a six-sided die. Before her throw, Bob bet that the outcome would be a 3 or a 4. We don't know the outcome, but we hear two reports. Mary says that the die landed on 2, 3, or 4; Sue says that it landed on 3, 4, or 5. We have no idea whether the reports are accurate.

- (26) If Mary's report is accurate or Sue's report is accurate,
probably Bob won.

Santorio observes that (26) has a true reading here. Compare with:

- (27) If the die landed 2, 3, or 4, probably Bob won.

We may parse the antecedent as $A \vee B$ or as $!(A \vee B)$.

It's time for dessert. Alice's parents play a game to decide what dessert she is allowed to have. They roll a die. If it lands 1 or 2, she is allowed fruit (and only fruit). If it lands 3 or 4, she is allowed ice cream (and only ice cream). If it lands 5 or 6, she is allowed cake (and only cake). The die has just been cast, but we don't know how it landed. What is the probability that Alice is allowed to have ice cream or cake?

A reasonable answer: two thirds. But no matter what happens, Alice is allowed exactly one dessert. There is zero chance that she is allowed ice cream *and* allowed cake.

Account of independence inferences

Recall the Canasta example (4). If you take any of the cards in the discard pile, the other cards are left they are.

This is determined by the conditional selection function in the usual way, without requiring any exhaustive interpretation. The alternatives are *independent* in the sense that supposing one alternative true leaves others false.

Similarly, the selection function may determine that if you take card A, B is left where it is, and if you take card B, A is left where it is.

In other cases, independence fails.

- If bishops A meets a bishop, the alternative 'Bishop B meets a bishop' may go from false to true—since A might be meeting B.
- If you go to the UK, among the possibilities we may consider is one where you go to Northern Ireland. In that case, the alternative 'You go to Ireland' goes from true to false.

Account of logically equivalent sentences

Some sentences may be logically equivalent but have different alternatives.

Multiple alternatives	Single alternative
Disjunction	Negation
<i>At least n</i>	<i>More than n</i>
<i>Any</i>	<i>Talk to each other, agree</i>
<i>Infinitely many</i>	<i>It snowed</i>

Table 1: Which items generate multiple alternatives?

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